Resampling Structured Data

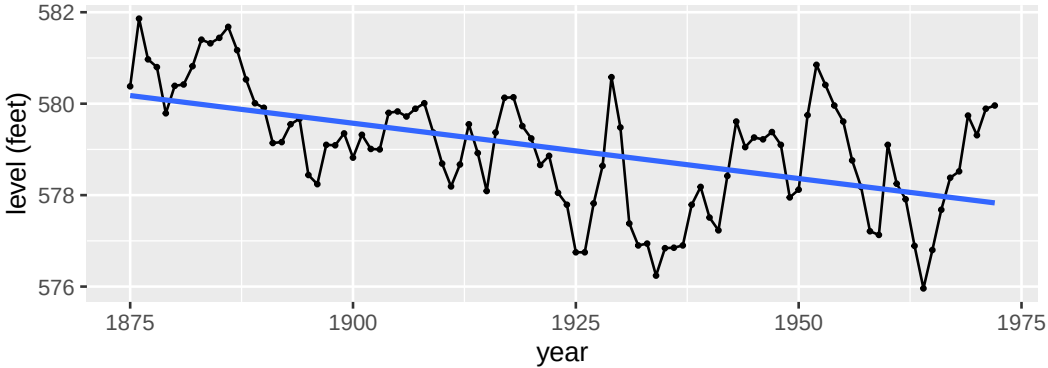
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Module 3: Resample

Computational Methods in Statistics and Data Science (DATASCI 406)

Is Lake Huron falling?

`LakeHuron` ships with R: the level of the lake in feet, measured once a year from 1875 to 1972.



Goal: confidence interval for the slope of the line.

The trend

```
fit_h <- lm(huron ~ year)
y_hat_h <- fitted(fit_h)
z <- resid(fit_h)
summary(fit_h)$coefficients["year", 1:2]
```

```
##   Estimate Std. Error
##  -0.024201   0.004036
```

```
slope <- function(y) cov(year, y) / var(year)
```

The lake fell about 2.4 feet per century. `slope` computes the least squares slope of any series against `year`.

The SE from `summary` assumes the residuals are independent. Is this a reasonable idea?

Limits of the bootstrap

So far the bootstrap has assumed **independent** and **identically distributed** observations.

- Levels in neighboring years may be dependent, through longer-term weather patterns or man-made changes.
- The distribution of water levels may change over the period.

Solution: use a model to say which parts of the data are independent, and resample those.

Models say what to resample

Model	Resample
X_t iid	observations (deck 1)
random walk, $X_t = X_{t-1} + \epsilon_t$	differences $X_t - X_{t-1}$
trend plus stationary noise, $X_t = m(t) + Z_t$	blocks of residuals
$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i$, ϵ_i iid	residuals
the same, with $\mathrm{Var}(\epsilon_i)$ depending on x_i	residuals with random signs
(X_i, Y_i) iid	rows
groups with their own distributions and fixed sizes	rows within each group

Dependent Data

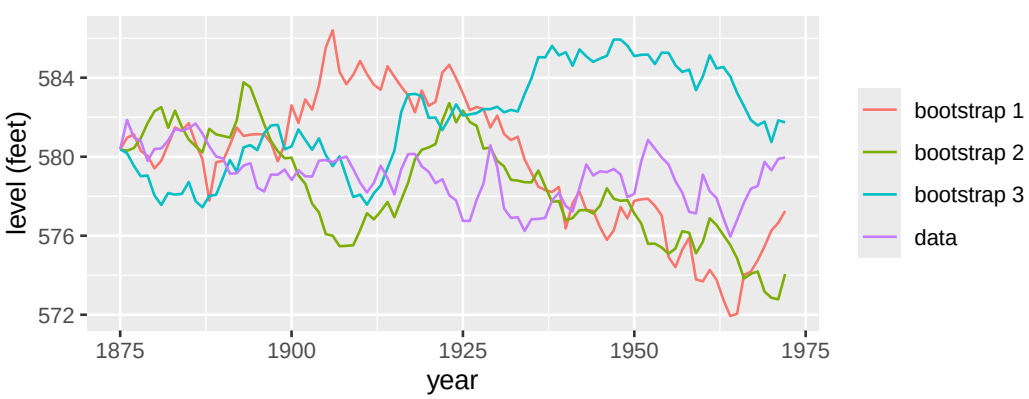
Model 1: a random walk

If $X_t = X_{t-1} + \epsilon_t$ with ϵ_t iid, the differences are iid. Resample them, and add them up from x_1.

```
d <- diff(huron)
boot_rw <- replicate(B, {
  d_star <- sample(d, replace = TRUE)
  slope(huron[1] + c(0, cumsum(d_star)))
})
c(random_walk = sd(boot_rw),
  summary = summary(fit_h)$coefficients["year", "Std. Error"])
```

```
## random_walk     summary
##    0.082465    0.004036
```

A random walk wanders



The bootstrap series drift far from 579 feet and stay there. Lake Huron's level keeps returning, which the differences alone cannot reproduce. The model is wrong, and its SE is 20 times the one from `summary`.

Model 2: a trend plus stationary noise

Write the series as

$$X_t = m(t) + Z_t,$$

with $m(t) = \beta_0 + \beta_1 t$ and Z_t **stationary**: its distribution does not change over time, so it has the same mean and variance in every year, and correlations that depend only on the lag.

To bootstrap the slope:

1. resample the residuals z_t,
2. add them to the fitted trend, $x_t^* = \hat{m}(t) + z_t^*$,
3. refit the slope.

Naive method: resample single residuals

```
boot_iid <- replicate(B, slope(y_hat_h + sample(z, replace = TRUE)))
c(bootstrap = sd(boot_iid),
  summary = summary(fit_h)$coefficients["year", "Std. Error"])
```

```
## bootstrap   summary
##  0.003919  0.004036
```

This is a regression bootstrap: `year` stays fixed, residuals are drawn one at a time and added to the fitted line, and the slope is refit. It agrees with `summary`, and both assume the residuals are independent.

Neighboring years are close

Pair each residual with the next year's, and correlate:

```
cor(z[-n], z[-1])
```

```
## [1] 0.7763
```

This is the **lag-1 autocorrelation**. The lag-j autocorrelation ρ_j pairs Z_t with Z_{t+j}, and `acf` computes it for several j:

```
round(acf(z, lag.max = 8, plot = FALSE)$acf[-1], 2)
```

```
## [1] 0.76 0.46 0.26 0.14 0.08 0.03 0.02 0.05
```

The correlations are near 0 by lag 5. Resampling runs of about that length keeps them.

Resample runs, not points

Fix a **block length** ℓ. The $n - \ell + 1$ blocks are the runs

$$(z_j, z_{j+1}, \ldots, z_{j+\ell-1}), \quad j = 1, \ldots, n - \ell + 1.$$

To draw one bootstrap series, draw blocks with replacement and put them end to end until there are n values. Keep the first n.

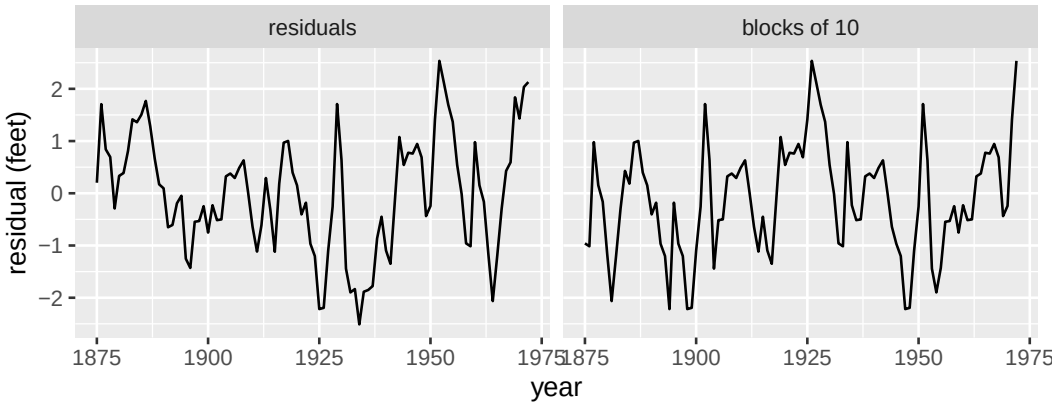
Within a block the dependence is kept. It is broken only where two blocks meet.

In R

```r
block_boot <- function(x, stat, B, l) {
  n <- length(x)
  replicate(B, {
    xs <- c()
    while (length(xs) < n) {
      i <- sample(n - l + 1, 1)
      xs <- c(xs, x[i:(i + l - 1)])
    }
    stat(xs[1:n])
  })
}
```

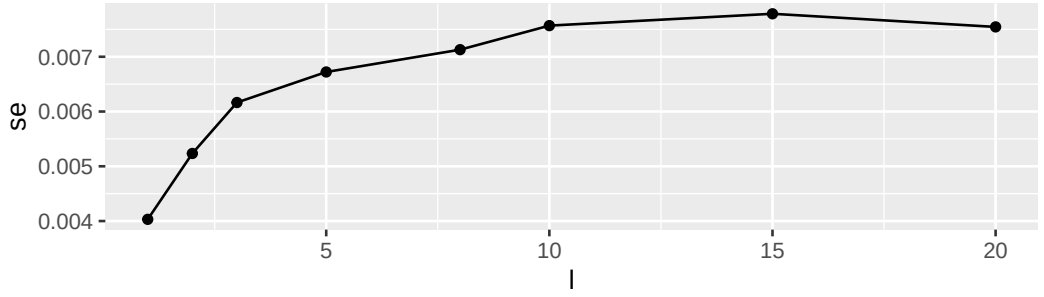
A block starts at most at $n - \ell + 1$, so that it ends by x_n.

A block bootstrap series



The SE of the slope against ℓ

```
ls <- c(1, 2, 3, 5, 8, 10, 15, 20)
boot_slope <- \(l) block_boot(z, \(zs) slope(y_hat_h + zs), B, l)
se_l <- map_dbl(ls, \(l) sd(boot_slope(l)))
ggplot(data.frame(l = ls, se = se_l), aes(l, se)) +
  geom_line() + geom_point()
```



Reading the plot

- $\ell = 1$ is the naive method.
- The SE rises with ℓ and levels off near 0.0075 for $\ell \geq 8$, about twice the SE from `summary`.
- A block of length ℓ keeps the correlations up to lag $\ell - 1$. Once ρ_j is near 0 beyond lag ℓ, longer blocks add nothing.

Choosing ℓ

- ℓ has to be **long** compared with the reach of the dependence, so a block keeps it, and **short** compared with n, so there are many blocks to draw.
- In practice, compute the SE for a range of ℓ and use a value where it has leveled off.
- In theory, the best ℓ for an SE grows like $n^{1/3}$ (Bühlmann, *Bootstraps for Time Series*, Statistical Science, 2002): 4.6 for $n = 98$. The constant depends on the dependence, so the rule gives a rate, not a value.

CI based on block bootstrap

```
se_10 <- sd(boot_slope(10))
coef(fit_h)[["year"]] + c(-1.96, 1.96) * se_10
```

```
## [1] -0.03903 -0.00937
```

Independent, Not Identically Distributed

Independent, not identical

Now suppose the observations are independent, but each has its own distribution:

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i,$$

so each Y_i has its own mean, and possibly its own variance.

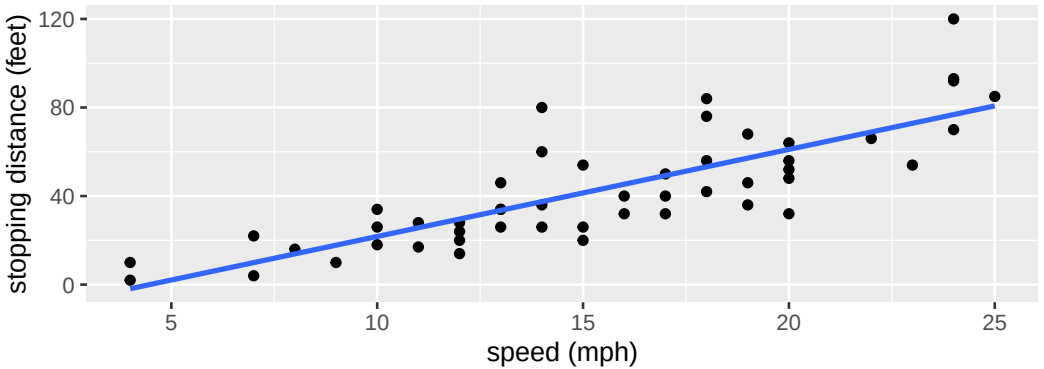
- **Resample cases**: draw rows (x_i, y_i) with replacement, as for the Flint pairs in deck 1.
- **Resample errors**: keep each x_i, and set

$$y_i^* = \hat{y}_i + e_i^*,$$

where $\hat{y}_i$ is the fitted value and $e_1^*, \ldots, e_n^*$ are drawn with replacement from the residuals $e_1, \ldots, e_n$. This is the naive method for Lake Huron, blocks of length 1.

Stopping distances

`cars` ships with R: the speed (mph) and stopping distance (feet) of 50 cars, recorded in the 1920s.



The fitted line

```
x <- cars$speed
y <- cars$dist
n_c <- length(x)
fit <- lm(dist ~ speed, data = cars)
y_hat <- fitted(fit)
e <- resid(fit)
coef(fit)
```

```
## (Intercept)       speed
##     -17.579       3.932
```

y_hat holds the fitted values $\hat{y}_i$, and e the residuals $e_i = y_i - \hat{y}_i$.

Resampling cases

Draw rows, and refit the line to each bootstrap sample. As for Lake Huron, `cov(x, y) / var(x)` is the least squares slope.

```
slope_cases <- replicate(B, {
  i <- sample(n_c, replace = TRUE)
  cov(x[i], y[i]) / var(x[i])
})
sd(slope_cases)
```

```
## [1] 0.3988
```

Resampling errors

Build each y^* from the fitted values and resampled residuals, and refit.

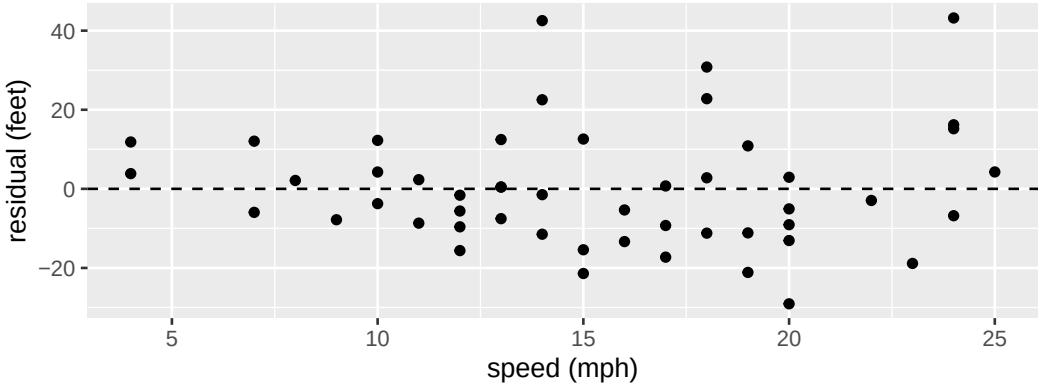
```
slope_errors <- replicate(B, {
  y_star <- y_hat + sample(e, replace = TRUE)
  cov(x, y_star) / var(x)
})
c(cases = sd(slope_cases), errors = sd(slope_errors),
  formula = summary(fit)$coefficients["speed", "Std. Error"])
```

```
##   cases  errors formula
##  0.3988  0.3987  0.4155
```

All three agree for `cars`. Resampling errors assumes more: the errors are iid and do not depend on x.

The Wild Bootstrap

Residuals against speed



The residuals spread out as speed increases.

Resampling errors moves residuals

A residual drawn for a car at 4 mph may come from a car at 24 mph.

- The cars at the extreme speeds have the most influence on the slope.
- If the errors are largest at high speeds, the real slope varies more than the bootstrap slopes, which spread the large residuals evenly over all speeds.
- Then resampling errors understates the SE.

Keep each residual where it is

The **wild bootstrap** keeps x fixed, and keeps each residual at its own x_i, with a random sign:

$$y_i^* = \hat{y}_i + e_i V_i, \qquad V_i = \begin{cases} +1 & \text{with probability } 1/2, \\ -1 & \text{with probability } 1/2. \end{cases}$$

- A residual of -30 feet at 24 mph becomes ± 30 feet at 24 mph.
- The x values are the original ones, as when resampling errors.

Why it works

Each y_i^* has mean $\hat{y}_i$ and variance e_i^2, so the bootstrap spread at each speed comes from the residual at that speed.

- Large residuals at high speeds stay at high speeds, where they move the slope most.
- One squared residual is a poor estimate of the error variance at its own x_i. But the slope combines all n of them, and the errors average out.

In R

```
slope_wild <- replicate(B, {
  V <- sample(c(-1, 1), n_c, replace = TRUE)
  cov(x, y_hat + e * V) / var(x)
})
c(cases = sd(slope_cases), errors = sd(slope_errors),
  wild = sd(slope_wild))
```

```
##  cases errors   wild
## 0.3988 0.3987 0.3977
```

For `cars` the three agree. The spread grows with speed, but not enough at the extreme speeds to change the SE.

Cases or wild?

- **Errors** assume the errors are iid and do not depend on x. Plot the residuals against x first.
- **Cases** allow random x. The fitted line does not have to be the true mean, since each bootstrap sample refits it.
- **Wild** keeps the original x values, so no bootstrap sample loses the extreme speeds. It needs the fitted line to be right, since every y_i^* is built from $\hat{y}_i$.

Strata

US states

`state.x77` ships with R: figures for the 50 states from the 1970s. `state.region` puts each state in one of four regions.

```
states <- data.frame(state.x77, region = state.region)
sapply(split(states$HS.Grad, states$region), median)
```

```
##     Northeast         South North Central          West
##         54.70         41.70         53.25         62.60
```

`HS.Grad` is the percent of adults with a high school diploma. The regions differ: the South is lowest and the West highest.

The interquartile range

How spread out are graduation rates across the states?

```
quantile(states$HS.Grad, c(0.25, 0.75))
```

```
##   25%   75%
## 48.05 59.15
```

```
IQR(states$HS.Grad)
```

```
## [1] 11.1
```

- The **interquartile range** is the 75th percentile minus the 25th: the width of the middle half of the states.
- It has no simple formula for its SE, so we bootstrap it.

Resample within each region

Each region has a fixed number of states. Treat the regions as strata, and resample states within each one.

```
in_region <- split(1:50, states$region)
iqr_hs <- \(i) IQR(states$HS.Grad[i])
boot_strata <- replicate(10000, {
  iqr_hs(unlist(map(in_region, \(k) sample(k, replace = TRUE))))
})
sd(boot_strata)
```

```
## [1] 2.352
```

Every bootstrap sample has 9, 16, 12, 13 states, as the data do.

Or resample all 50 states?

```
boot_rows <- replicate(10000, iqr_hs(sample(50, replace = TRUE)))
c(strata = sd(boot_strata), all_rows = sd(boot_rows))
```

```
##   strata all_rows
##    2.352    2.997
```

- Resampling all 50 lets the number of states from each region vary. One sample might have 12 Southern states, the next 20.
- The South is low and the West high, so a changing mix changes the spread, and the SE is 27% larger.
- That extra variation is not in the data: the regions always have 9, 16, 12, 13 states.

Comparing groups

How much higher is the West's median graduation rate than the South's?

- A statistic that compares groups computes one piece from each group, here a median for each region.
- Resampling within each region computes every median from the region's own number of states: 13 Western and 16 Southern, every time.
- Resampling all 50, 95% of samples would have between 7 and 19 Western states, and between 10 and 23 Southern.

The West minus the South

```
west_south <- function(i) {
  r <- states$region[i]
  g <- states$HS.Grad[i]
  median(g[r == "West"]) - median(g[r == "South"])
}
boot_ws <- replicate(10000, {
  west_south(unlist(map(in_region, \(k) sample(k, replace = TRUE))))
})
c(estimate = west_south(1:50), quantile(boot_ws, c(0.025, 0.975)))
```

```
## estimate     2.5%    97.5%
##     20.9     12.4     22.9
```

Summary

Summary

- Each scheme starts from a model that says what is iid, and resamples that.
- A random walk makes the differences iid. It does not fit a series that keeps returning to its level, such as Lake Huron.
- The **moving block bootstrap** resamples runs of length ℓ from a **stationary** series. Remove a trend first, resample the residuals in blocks, and add the trend back. Increase ℓ until the SE levels off.
- In regression, **resample cases**, or **resample errors** when the error variance does not depend on x.
- The **wild bootstrap** keeps each residual at its own x_i and flips its sign at random. It allows the error variance to change with x.
- When groups have fixed sizes, or a statistic compares groups, **resample within each group**.